The Lie Ring of Analytic Functions on the Disk is Algebraically Determined

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Definition

By a **Polish space** we mean a separable topological space that admits a compatible complete metric.

Definition

By a **Lie ring** we mean an Abelian group $\mathcal{L}$ with an additional operation $[\cdot,\cdot] : \mathcal{L}^2 \to \mathcal{L}$ so that

- $[x, x] = 0$;
- $[x + y, z] = [x, z] + [y, z]$;
- $[x, y + z] = [x, y] + [x, z]$;
- $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$.

We say that $\mathcal{M} \subseteq \mathcal{L}$ is a **Lie subring** if $\mathcal{M}$ is a subgroup and

$$(\forall x, y \in \mathcal{M})([x, y] \in \mathcal{M}).$$

Definition

By a **Polish Lie ring** we mean a Lie ring $\mathcal{L}$ endowed with a Polish topology so that the following are continuous:

- $(x, y) \mapsto x + y$, $\mathcal{L}^2 \to \mathcal{L}$,
- $x \mapsto -x$, $\mathcal{L} \to \mathcal{L}$,
- $(x, y) \mapsto [x, y]$, $\mathcal{L}^2 \to \mathcal{L}$

Definition

Let $\mathcal{L}$ be a Polish Lie ring. We say that $\mathcal{L}$ is **algebraically determined** provided that, for any Polish Lie ring $\mathcal{M}$, if $\varphi : \mathcal{M} \to \mathcal{L}$ is an isomorphism of Lie rings, φ is a homeomorphism.

Recall the well-known theorem.

Theorem

Let $\varphi : G \to H$ be a group homomorphism between Polish groups G and H. If φ is Baire-measurable, φ is continuous.

Using the fact that continuous injective images of Borel sets are Borel when mapping between Polish spaces, we see that any Baire-measurable group isomophism between Polish groups is, in fact, a homeomorphism.
In [1], Dr. Kallman showed that S_∞ has only one Polish group topology.

Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ and let $\mathcal{A}(\mathbb{D})$ be the collection of complex analytic functions on $\mathbb{D}$. We define $[f, g] = fg' - f'g$, $\mathcal{A}(\mathbb{D})^2 \to \mathcal{A}(\mathbb{D})$, where h' is the derivative of h.

Theorem

Let G be a Polish group and $\varphi : G \to \mathcal{A}(\mathbb{D})$ be an additive group isomorphism. If there exists $\{\lambda_n : n \in \omega\} \subseteq \mathbb{D}$ dense in $\mathbb{D}$ with the property that, for each $n \in \omega$, $x \mapsto \varphi(x)(\lambda_n)$, $G \to \mathbb{C}$, is Baire-measurable, then φ is a homeomorphism.

For the remainder, let $\mathcal{L}$ be a Polish Lie ring and $\varphi : \mathcal{L} \to \mathcal{A}(\mathbb{D})$ be an isomorphism of Lie rings.

Lemma

$\varphi^{-1}[\mathbb{C}]$ is a closed Lie subring of $\mathcal{L}$.

Proof.

This follows from the observation that

$$\varphi^{-1}[\mathbb{C}] = \{x \in \mathcal{L} : [\varphi^{-1}(1), x] = 0\}.$$

Since the bracket is continuous, we see that $\varphi^{-1}[\mathbb{C}]$ is closed. $\square$

For $a \in \mathbb{D}$, let

$$\mathcal{M}_a = \{\langle f, G, \lambda \rangle \in \mathcal{L}^2 \oplus \varphi^{-1}[\mathbb{C}] : f = \lambda + G + [\varphi^{-1}(z-a), G]\}.$$

and

$$C_a = \{0\} \oplus \varphi^{-1}[\mathbb{C}] \oplus \{0\}.$$

Notice that both $\mathcal{M}_a$ and C_a are closed additive subgroups of $\mathcal{L}^3$. To see that $C_a \subseteq \mathcal{M}_a$, notice that

$$\varphi(\varphi^{-1}(\lambda) + [\varphi^{-1}(z-a), \varphi^{-1}(\lambda)]) = \lambda - \lambda = 0.$$

Hence, C_a is a closed subgroup of $\mathcal{M}_a$. By [2, Theorem 12.17], we can find $B \subseteq \mathcal{M}_a$, a Borel transversal of $\sim_{C_a}$.

Let $\pi_1 : \mathcal{L}^3 \to \mathcal{L}$ be the canonical projection onto the first coordinate.

Lemma

$\pi_1 \restriction_B$ is a bijection.

Lemma

For $a \in \mathbb{D}$, the map $x \mapsto \varphi^{-1}(\varphi(x)(a))$, $\mathcal{L} \to \varphi^{-1}[\mathbb{C}]$, is a Borel mapping.

Let $\mathcal{B} = \{\mathbb{D} + k : k \in \mathbb{Q}^2\}$. The family $\mathcal{B}$ forms a subbase for the topology on $\mathbb{C}$.

Lemma

$\varphi^{-1}[\mathbb{D}]$ *is co-analytic.*

Proof.

Recall that $\varphi^{-1}[\mathbb{C}]$ is closed. We show that $\varphi^{-1}[\mathbb{C} \setminus \mathbb{D}]$ is analytic. This follows from the fact that $x \in \varphi^{-1}[\mathbb{C} \setminus \mathbb{D}]$ if and only if

$$(\exists G \in \mathcal{L})([\varphi^{-1}(z) - x, G] + G - \varphi^{-1}(1) = 0).$$

□

Lemma

$\varphi \restriction_{\varphi^{-1}[\mathbb{C}]}$ is Baire-measurable.

Proof.

This follows from the fact that $\mathcal{B}$ is a subbase for the topology on $\mathbb{C}$ and that each member of $\mathcal{B}$ is co-analytic, hence has the Baire property by Luzin-Sierpiński. $\square$

Theorem

The Polish Lie ring $\mathcal{A}(\mathbb{D})$ is algebraically determined.

Proof.

Apply the previous lemma and the theorem about point evaluation. $\square$

[1] Robert R. Kallman, *A uniqueness result for the infinite symmetric group*, Studies in Analysis, Adv. in Math. Suppl. Studies **4** (1979), 321–322.

[2] Alexander Kechris, *Classical descriptive set theory*, Springer-Verlag, 1995.